



Smart greenhouse

Presented to: Dr. Mohamad Aoude.

Presented by :

Fatima jaafar (6591)

Mohamad injibar (6915)

Mostafa alawiyeh (6884)

Razan najem (6694)

1. Introduction	2
2. Problem Statement.....	2
3. System Architecture	3
3.1 Overall System Architecture	3
1) Sensing Layer (Input Sensors).....	3
2) Processing and Decision Layer	4
3) Actuation Layer (Outputs)	4
4) Monitoring and Security Layer	4
4. Sensor Selection and Justification.....	5
4.1 DHT11 Temperature and Humidity Sensor	5
Sensor Description	5
Justification and System Role	5
4.2 Soil Moisture Sensor	6
Sensor Description	6
Justification and System Role	6
4.3 LDR Light Sensor	6
Sensor Description	6
Justification and System Role	6
4.4 Water Level Sensor	7
Sensor Description	7
Justification and System Role	7
4.5 Raindrop Sensor	7
Sensor Description	7
Justification and System Role	7
4.6 Ultrasonic Sensor (HC-SR04).....	8
Sensor Description	8
Justification and System Role	8
4.7 IR Receiver Module	8
Sensor Description	8
Justification and System Role	8
5. Hardware Design and Schematics	9
5.1 Hardware Overview.....	9

5.2 Arduino Mega 2560 Controller	9
5.3 Sensor Connections	10
Temperature and Humidity Sensor (DHT11)	10
Soil Moisture Sensor	10
Light Sensor (LDR).....	10
Water Level Sensor	10
Raindrop Sensor	10
Ultrasonic Sensor HC-SR04	10
5.4 Actuator Connections	11
Water Pump	11
Cooling Fan Motor.....	11
Servo Motor SG90	11
5.5 User Interface and Security Components	11
I2C LCD Display	12
LED Indicators and Buzzer	12
5.6 Power Supply Design.....	12
Low Power Circuit	12
6. Software Design and Firmware Architecture	13
6.1 Firmware Overview	13
The software of the smart greenhouse system was developed using Arduino C/C++ and implemented on the Arduino Mega 2560 platform. The program is designed to manage sensor data acquisition, automatic environmental control, user interaction, and security functions.....	13
6.2 Software Initialization (setup function)	13
6.3 Main Program Execution (loop function)	14
6.4 Sensor Data Acquisition Functions.....	14
Temperature and Humidity Measurement	14
Soil Moisture Measurement	15
Light Measurement	15
Rain Detection	15
Water Level Measurement	16
Distance Measurement	16

6.5 Automatic Control Algorithms.....	16
Fan Control Algorithm	16
Irrigation Control Algorithm.....	17
Roof Servo Control Algorithm	17
Roof Opening:.....	17
Roof Closing:.....	18
6.6 Operating Modes of the System	18
Mode 0: Locked Mode	18
Mode 1: Access Granted Mode.....	18
Mode 2: Monitoring Mode	19
Mode 3: Alarm Mode	19
6.7 IR Remote Control Software.....	19
6.8 LCD Display Software.....	20
6.9 Serial Monitoring.....	20
7. Calibration Procedures and Results	21
7.1 Calibration Overview.....	21
7.2 DHT11 Temperature and Humidity Sensor Calibration	21
Calibration Method	21
Calibration Procedure	21
Result	22
7.3 Soil Moisture Sensor Calibration	22
Calibration Method	22
Calibration Procedure	22
System Application	22
7.4 LDR Light Sensor Calibration	23
Calibration Method	23
Calibration Procedure	23
System Application	23
7.5 Water Level Sensor Calibration	23
Calibration Method	23
Calibration Levels	23
7.6 Raindrop Sensor Calibration	24

Calibration Method	24
Calibration Procedure	24
System Application	24
7.7 Ultrasonic Sensor Calibration	25
Calibration Method	25
Calibration Table	25
System Application	25
7.8 Calibration Summary	25
8. Experimental Results and Data Analysis	25
8.1 System Testing Procedure	26
8.2 Temperature Control Test.....	26
Objective.....	27
Test Procedure	27
Expected Behavior	27
Result	27
8.3 Automatic Irrigation Test	27
Objective.....	27
Test Procedure	27
Expected Behavior	27
8.4 Greenhouse Roof Control Test	28
Objective.....	28
Test Conditions.....	28
System Logic	28
Result	28
8.5 Security System Test	29
Objective.....	29
Test Procedure	29
System Response	29
Result	29
8.6 LCD Monitoring Test.....	29
Page 1:.....	30
Page 2:.....	30

Page 3:.....	30
Page 4:.....	30
8.7 Serial Data Monitoring.....	30
8.8 Experimental Results Summary	31
9. Measurement Uncertainty and Error Analysis.....	31
9.1 Introduction to Measurement Uncertainty	31
9.2 Sensor Accuracy and Resolution.....	31
DHT11 Temperature and Humidity Sensor	32
Soil Moisture Sensor	32
LDR Light Sensor.....	32
Water Level Sensor	33
Raindrop Sensor	33
Ultrasonic Sensor	33
9.3 Sources of Measurement Noise	34
9.4 Repeatability Analysis	34
9.5 Error Reduction and Possible Improvements	34
9.6 Summary	35
10. Engineering Constraints Discussion	35
10.1 Cost Considerations	35
10.2 Reliability and Robustness	35
10.3 Electrical Safety.....	36
10.4 Power Consumption and Management.....	36
Low-Power Components	36
High-Power Components	37
10.5 Maintainability.....	37
10.6 Scalability and Future Improvements	38
10.7 Summary	38
11. Conclusion and Future Work	38
11.1 Conclusion.....	39
11.2 Future Work	39
1. IoT-Based Remote Monitoring	39
2. Data Storage and Analysis.....	40

3. Improved Sensor Accuracy.....	40
4. Intelligent Control Algorithms.....	40
5. Solar Power Integration.....	40
6. Expansion of Automation Functions.....	40
11.3 Final Remarks.....	41
References	41

1. Introduction

Agriculture is facing increasing challenges due to limited resources, climate changes, and the need for higher productivity. Traditional farming methods require continuous human monitoring and may lead to inefficient use of water, energy, and other resources.

A smart greenhouse provides an intelligent solution by using sensors and automation technologies to monitor and control environmental conditions. This project presents an Arduino-based smart greenhouse system that integrates different sensors to measure important parameters such as light intensity, temperature, humidity, water flow, and plant-related conditions.

The collected data are processed to evaluate greenhouse conditions, detect abnormal situations, and trigger automatic responses when needed. The objective of this project is to apply sensors and instrumentation concepts to develop a reliable monitoring system that improves plant growth conditions and optimizes resource usage.



2. Problem Statement

Maintaining suitable environmental conditions is one of the main challenges in greenhouse agriculture. Traditional greenhouse systems usually require continuous human observation to monitor parameters such as temperature, humidity, soil moisture, water availability, and lighting conditions. This manual monitoring approach can lead to delayed responses, inefficient resource consumption, and unfavorable conditions that may affect plant growth.

The main problem addressed in this project is the need for an intelligent monitoring and control system capable of continuously measuring greenhouse conditions and automatically reacting to environmental changes. Variations in temperature, humidity,

soil moisture, water level, and light intensity can negatively influence plant health if they are not detected and corrected in time.

To solve this problem, an Arduino Mega-based smart greenhouse system is proposed. The system integrates multiple sensors to collect real-time environmental data and uses this information to control different actuators, including a ventilation fan, irrigation pump, and servo mechanism. Warning indicators such as LEDs and a buzzer are also included to notify users when abnormal conditions are detected.

The proposed system aims to reduce human intervention, optimize water and energy consumption, and provide a reliable platform for monitoring and automatically controlling greenhouse conditions.

3. System Architecture

3.1 Overall System Architecture

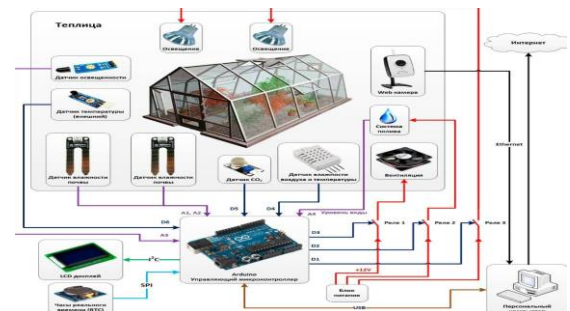
The proposed smart greenhouse system is based on an **Arduino Mega microcontroller** that acts as the main processing and control unit. The Arduino Mega collects data from multiple sensors, processes the measurements according to predefined thresholds, and generates appropriate control signals for actuators and user notification devices.

The system consists of four main layers:

1. **Sensing Layer**
2. **Processing and Decision Layer**
3. **Actuation Layer**
4. **Monitoring and Security Layer**

1) Sensing Layer (Input Sensors)

The Arduino Mega continuously receives environmental information from different sensors:



Sensor	Function	Arduino Input
DHT11	Measures temperature and humidity	Digital pin 13
Soil Moisture Sensor	Measures soil water content	Analog pin A1
Water Level Sensor	Monitors water reservoir level	Analog pin A8
LDR Sensor	Measures light intensity	Analog pin A12
Raindrop Sensor	Detects rain intensity	Analog pin A10
Ultrasonic Sensor (HC-SR04)	Detects intrusion/object distance	Trig pin 4 / Echo pin 3
IR Receiver	Password/security control	Digital pin 2

2) Processing and Decision Layer

The **Arduino Mega 2560** is the central controller of the system.

Its main functions are:

- Reading all sensor values.
- Processing analog and digital signals.
- Comparing measurements with predefined thresholds.
- Taking automatic decisions.
- Controlling actuators.
- Sending information to the LCD display.

The implemented control logic includes:

- If temperature > 31°C → activate ventilation fan.
- If soil moisture < 35% → activate irrigation pump.
- If light intensity is high and no storm is detected → open roof using servo motor.
- If rain intensity is high → close roof.
- If unauthorized movement is detected → activate alarm system.

3) Actuation Layer (Outputs)

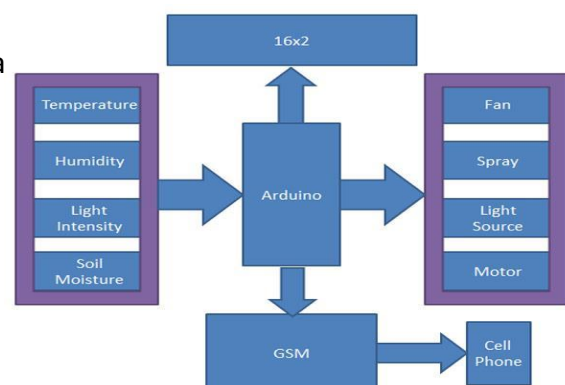
Based on sensor readings, the Arduino controls different actuators:

Actuator	Function	Control
DC Fan Motor (3–6V)	Greenhouse ventilation	Relay
Water Pump (5V)	Automatic irrigation	Relay
Servo Motor SG90	Roof opening/closing mechanism	PWM signal
Red LED	Warning indicator	Digital output
Green LED	Normal operation indicator	Digital output
Buzzer	Alarm notification	Digital output

4) Monitoring and Security Layer

The system provides user interaction and protection through:

- **I2C LCD 16×2 display**
 - Temperature and humidity display
 - Water level monitoring.
 - Light and rain information.
 - Soil moisture percentage.
- **IR Remote Security System**
 - Allows password entry.
 - Prevents unauthorized access.



- **Ultrasonic Intrusion Detection**

- Detects objects/persons near the greenhouse.
- Activates red LED and buzzer alarm when intrusion is detected.

4. Sensor Selection and Justification

The smart greenhouse system requires continuous monitoring of environmental parameters to maintain suitable conditions for plant growth and ensure safe operation. Several sensors were selected to collect real-time measurements, which are processed by the Arduino Mega controller to perform automatic decisions. Each sensor was chosen according to its measurement capability, compatibility with the Arduino platform, and its importance in greenhouse automation.

4.1 DHT11 Temperature and Humidity Sensor

Sensor Description

The DHT11 sensor was selected to measure the ambient temperature and relative humidity inside the greenhouse. These two parameters are essential because they directly influence plant growth, water evaporation, and overall greenhouse climate.

The sensor provides digital output data, which allows direct communication with the Arduino Mega without requiring additional signal conditioning.



Justification and System Role

The DHT11 sensor serves as the primary environmental feedback trigger for thermodynamic regulation within the greenhouse. Temperature management is crucial for plants to regulate transpiration rates, stomatal opening, and photosynthetic efficiency; prolonged exposure to temperatures above optimal biological thresholds can lead to heat stress, enzyme denaturation, and wilting.

- **System Role & Threshold Logic:** When the ambient temperature surpasses 31 °C, the system initiates active ventilation by switching on the cooling fan relay. This active airflow expels trapped hot air and draws in cooler outside air. Once the internal temperature cools down to or below 31 °C, the fan is turned off to prevent unnecessary power consumption and over-cooling.
- **Engineering Note on Demonstration Thresholds:** While 31 °C is programmed as the activation setpoint, this specific value is chosen for rapid testing and demonstration purposes under indoor laboratory conditions. In a practical

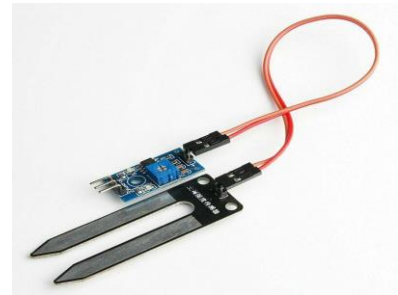
deployment, this boundary would be dynamic or set lower (e.g., 24 °C - 28°C) to match the specific botanical and seasonal requirements of the target crop, demonstrating the scalability of our control algorithms.

4.2 Soil Moisture Sensor

Sensor Description

The soil moisture sensor was selected to measure the amount of water present in the soil. This parameter is essential for determining irrigation requirements and avoiding both plant dehydration and excessive watering.

The sensor provides an analog signal proportional to the moisture level, which is converted by the Arduino Mega ADC into a percentage value.



Justification and System Role

Justification and System Role: The soil moisture sensor provides closed-loop feedback to prevent both soil dehydration (which limits nutrient absorption and stunts plant growth) and soil waterlogging (which suffocates root structures and promotes fungal pathogens like root rot).

- **System Role & Threshold Logic:** To automate irrigation safely, the Arduino continuously polls soil moisture levels. When the mapped sensor percentage drops below 35%, it indicates that the root zone is reaching a critical drying state, prompting the Arduino to engage the 5V water pump relay. Crucially, the system continues to poll moisture levels *during* watering; once the sensor registers that the soil moisture has crossed the 35% threshold (plus a small software hysteresis gap), the pump automatically shuts off. This real-time loop prevents over-saturation and drastically reduces water waste.
- **Engineering Note on Demonstration Thresholds:** The 35% trigger threshold used in our demonstration program is selected specifically to allow for easy, visible state-transitions during lab testing (for instance, demonstrating a clear difference between an exposed sensor in air vs. a submerged sensor in a cup of soil). For real agricultural application, this threshold would be carefully calibrated based on the soil type (sandy, clayey, or loamy) and the specific permanent wilting point (PWP) of the cultivated crops.

4.3 LDR Light Sensor

Sensor Description

A Light Dependent Resistor (LDR) was selected to measure the intensity of light inside the greenhouse. The resistance of the LDR changes according to the amount of incident light, producing an analog voltage variation.

Photoresistor (LDR) Sensor:

In this project, a photoresistor was used to measure light intensity. Since a photoresistor is commonly referred to as an LDR (Light Dependent Resistor) and operates on the same principle, it is referred to as an LDR sensor throughout this report.



To convert the variable resistance of the LDR into a readable analog voltage, the sensor is configured as part of a static voltage divider circuit. Instead of an adjustable potentiometer, which can introduce mechanical instability, calibration drift, and contact resistance oxidation in a high-humidity greenhouse, a fixed, high-precision 10Kohms pull-down resistor was utilized. This design establishes a highly stable passive transducer circuit that outputs a reliable analog voltage proportional to the ambient light intensity.

Justification and System Role

The LDR sensor quantifies ambient sunlight levels, permitting the greenhouse to transition between passive solar heating and crop protection modes.

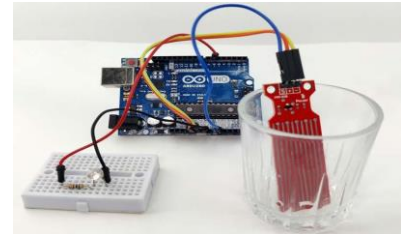
- **System Role & Threshold Logic:** Photosynthesis requires adequate solar radiation, yet structural heat build-up can occur under direct intensity. The roof control algorithm opens the ventilation hatches via the servo motor when ambient light exceeds 80%—provided no concurrent rain is detected—allowing maximum natural sunlight and fresh air circulation to reach the plants. Conversely, during low-light scenarios (below 80%) or night time, the roof is automatically closed. This locks in the internal thermal energy accumulated during the day, maintaining a warmer microclimate during colder night cycles.
- **Engineering Note on Demonstration Thresholds:** The high threshold of 80% is programmed to guarantee distinct control transitions during indoor laboratory testing, where artificial flashlights or desk lamps are used to simulate daylight. In an actual field installation, the threshold would be adjusted to account for natural lux levels of standard daylight cycles (typically opening the roof at much lower light levels like 20%-30% to maximize daily solar capture).

4.4 Water Level Sensor

Sensor Description

The water level sensor was selected to monitor the amount of water available in the irrigation reservoir. Maintaining sufficient water availability is important to ensure continuous operation of the irrigation system.

The sensor provides an analog signal corresponding to the detected water level.



Justification and System Role

The water level sensor serves a dual role: providing real-time resource telemetry to the operator and establishing a critical safety interlock for the mechanical system.

- **System Role & Safety Logic:** Running a standard centrifugal DC water pump when the reservoir is entirely dry causes immediate motor overheating, frictional wear, and permanent damage. The Arduino continuously reads the analog output from the level sensor and categorizes the reservoir status into four states: **DRY**, **LOW**, **MED**, and **HIGH**. While this is presented to the user on the LCD screen for manual resource replenishment, the firmware utilizes this state matrix to temporarily disable the irrigation pump relay if a **DRY** state is detected, overriding any automatic watering commands from the soil moisture sensor.
- **Engineering Note on Demonstration Thresholds:** The level boundaries for the four states are calibrated based on the dimensions of our prototype testing reservoir (a shallow cup or small container). In a scaled agricultural reservoir or rain-harvesting tank, the sensor values would be mapped to volumetric measurements (e.g., liters remaining) using a deep-well pressure transducer or non-contact ultrasonic depth sensor, rather than the contact-based conductive strip used in our lab prototype.

4.5 Raindrop Sensor

Sensor Description

The raindrop sensor was selected to detect rain occurrence and protect the greenhouse from unfavorable weather conditions.

The sensor generates an analog signal depending on the amount of water detected on its surface.



Justification and System Role

The raindrop sensor acts as a critical meteorological sensor, protecting the greenhouse's internal environment from physical damage and water logging.

- **System Role & Threshold Logic:** Uncontrolled rainwater entering a precision greenhouse can cause soil erosion, over-saturation of nutrients, damage to sensitive electronics, and elevated internal relative humidity that encourages mold growth. The sensor continuously measures surface conductivity. When the mapped precipitation intensity exceeds the storm threshold of 70%, the Arduino immediately commands the SG90 roof servo to close the roof. This storm logic takes absolute priority over the LDR's light-based roof-opening commands, ensuring structural protection is maintained during sudden storms even if ambient light remains bright.
- **Engineering Note on Demonstration Thresholds:** The storm threshold (70%) is set high in our program to ensure that small droplets of moisture or condensation on the sensor surface do not trigger accidental roof closures during testing. For rapid lab demonstration, applying a few drops of water instantly breaches this threshold. In a permanent installation, a rain sensor heater circuit would be included to evaporate residual moisture, and the threshold would be calibrated to distinguish between a light mist and actual downpours.

4.6 Ultrasonic Sensor (HC-SR04)

Sensor Description

The ultrasonic sensor was selected as a security element to detect unauthorized movement near the greenhouse.

It operates by transmitting ultrasonic waves and measuring the time required for the echo signal to return. The distance is then calculated by the Arduino.



Justification and System Role

The HC-SR04 ultrasonic sensor provides non-contact proximity detection, establishing a digital perimeter fence around the greenhouse structure.

- **System Role & Security Logic:** Agricultural assets and valuable crops inside greenhouses are susceptible to vandalism, theft, or animal intrusions. To safeguard the perimeter, the sensor emits high-frequency 40 kHz sonic pulses

and calculates target distance based on the time-of-flight of the returning echo. If a physical object breaches the protective envelope (defined as a distance of less than 11cm), the Arduino triggers an immediate transition into **Alarm Mode (Mode 3)**. This sounds the buzzer, flashes the red indicator, and locks the screen to request password verification. This automatic response acts as both an active deterrent and an operator alert system.

- **Engineering Note on Demonstration Thresholds:** A threshold of 11 cm was chosen specifically to match the physical dimensions of our tabletop prototype scale. This small distance allows a demonstrator to easily trigger the security system by placing their hand close to the prototype on a desk. In a commercial system, this ultrasonic sensor would be replaced by outdoor passive infrared (PIR) motion sensors or a long-range lidar/ultrasonic sensor array with a protective perimeter threshold mapped to several meters.

4.7 IR Receiver Module

Sensor Description

The IR receiver module was selected to provide wireless user interaction and security control.

It receives infrared signals transmitted by a remote control and sends decoded commands to the Arduino Mega.



Justification and System Role

The IR receiver module functions as the localized human-machine interface (HMI) for access control and alarm management.

- **System Role & Access Logic:** A secure automated system must provide an authenticated bypass mechanism for authorized operators to perform maintenance or harvesting without triggering alarms. The IR module decodes incoming 38 kHz modulated infrared pulses from the remote. The system forces the operator to enter a multi-character passcode (#123). If verified, the microcontroller shifts the state machine out of **Locked Mode (Mode 0)** or **Alarm Mode (Mode 3)** and into **Monitoring Mode (Mode 2)**, disengaging active alarms and restoring standard telemetry displays. If the passcode is entered incorrectly, access is denied, protecting the environmental configurations and physical actuators from unauthorized modification.
- **Engineering Note on Demonstration Thresholds:** The default passcode (#123) is hardcoded for simplicity and quick execution during academic evaluations

and grading. In an industry-standard deployment, a more secure cryptographic key or a rolling-code protocol would be implemented, and the passcode would be stored in the non-volatile EEPROM of the microcontroller, allowing the greenhouse administrator to update the password dynamically via a secure keypad or network interface.

5. Hardware Design and Schematics

5.1 Hardware Overview

The hardware implementation of the smart greenhouse system is based on an **Arduino Mega 2560** microcontroller, which acts as the main control unit responsible for acquiring sensor data, processing measurements, and controlling different actuators according to predefined conditions.

The system integrates several input sensors, output devices, and power management components to achieve automatic greenhouse monitoring and control. The complete electrical connection between the components is illustrated in Figure 9.

5.2 Arduino Mega 2560 Controller

The Arduino Mega 2560 was selected as the main processing unit because it provides a large number of digital and analog input/output pins, allowing the connection of multiple sensors and actuators simultaneously.

The Arduino performs the following tasks:

- Reading environmental parameters from sensors.
- Processing analog and digital signals.
- Applying control decisions according to programmed thresholds.
- Driving actuators through relay modules.
- Updating the LCD display with real-time measurements.
- Managing the security system using the IR receiver and ultrasonic sensor.

5.3 Sensor Connections

The system includes several sensors connected directly to the Arduino Mega:

Temperature and Humidity Sensor (DHT11)

The DHT11 sensor is connected to digital pin **D13** and provides temperature and humidity measurements. These values are processed by the Arduino to monitor greenhouse climate conditions.

Soil Moisture Sensor

The soil moisture sensor is connected to analog pin **A1**. It provides an analog signal proportional to the water content of the soil.

The Arduino uses this measurement to control irrigation:

- If soil moisture becomes lower than **35%**, the irrigation pump is activated.
- When the soil moisture reaches an acceptable level, the pump is switched OFF.

Light Sensor (LDR)

The LDR is integrated into a passive voltage divider network connected to analog pin A12. The circuit consists of the LDR in series with a fixed 10 Kohms resistor tied to Ground (GND). As ambient light levels increase, the resistance of the LDR drops significantly. According to the voltage divider equation:

$$V_{out} = V_{in} \cdot \left(\frac{R_{fixed}}{R_{LDR} + R_{fixed}} \right)$$

This drop in LDR resistance causes the analog output voltage (V_{out}) across the fixed resistor to rise towards 5V, which is then read by the Arduino Mega's ADC. This static design ensures excellent thermal stability and eliminates physical wear over long-term operation. The measured light intensity percentage is used to automate the roof mechanism: if the light intensity is higher than 80% and no heavy rain is detected, the servo opens the roof; otherwise, it remains closed.

Water Level Sensor

The water level sensor is connected to analog pin **A8**. It monitors the amount of water available in the reservoir and displays the status on the LCD.

Raindrop Sensor

The raindrop sensor is connected to analog pin **A10**. It detects rain intensity and protects the greenhouse by closing the roof during storm conditions.

Ultrasonic Sensor HC-SR04

The ultrasonic sensor uses:

- Trigger pin → D4
- Echo pin → D3

It is used for security monitoring by detecting nearby objects or unauthorized access.

5.4 Actuator Connections

Water Pump

The 5V water pump is controlled using a relay module connected to Arduino pin **D22**.

The relay provides electrical isolation between the Arduino control circuit and the pump power circuit.

Automatic operation:

- Dry soil detected → relay activated → pump ON.
- Required moisture reached → relay deactivated → pump OFF.

Cooling Fan Motor

The DC fan motor is controlled through a relay connected to Arduino pin **D52**.

Automatic operation:

- Temperature > 31°C → fan activated.
- Temperature ≤ 31°C → fan stopped.

The fan helps maintain a suitable greenhouse temperature.

Servo Motor SG90

The servo motor is connected to digital pin **D48**.

It controls the greenhouse roof mechanism:

- Roof opens during suitable weather conditions.
- Roof closes during rain or insufficient light.

5.5 User Interface and Security Components

I2C LCD Display

The 16×2 I2C LCD is used to display:

- Temperature
- Humidity
- Soil moisture percentage
- Water level status
- Light intensity
- Rain intensity

The LCD allows the user to monitor system status in real time.

LED Indicators and Buzzer

The system contains:

- Green LED → normal operation indication.
- Red LED → alarm condition.
- Buzzer → security warning signal.

When intrusion is detected by the ultrasonic sensor:

- Red LED starts blinking.
- Buzzer generates an alarm.
- The system requests password verification.

5.6 Power Supply Design

The power system is divided into two separate parts to ensure stable operation.

Low Power Circuit

The following components are supplied from the Arduino 5V VCC pin:

- Sensors
- LCD display
- LEDs
- Buzzer

The Arduino is powered through USB connection with the computer during operation.

Actuator Power Circuit

Because motors and pumps require higher current, they are powered separately using:

- Two Li-ion batteries connected in series (3000 mAh).
- Two step-down voltage regulators.
- Regulated 5V output for actuators.

This separation prevents voltage drops and protects the Arduino from current overload caused by inductive loads such as motors and pumps.

6. Software Design and Firmware Architecture

6.1 Firmware Overview

The software of the smart greenhouse system was developed using **Arduino C/C++** and implemented on the Arduino Mega 2560 platform. The program is designed to manage sensor data acquisition, automatic environmental control, user interaction, and security functions.

The firmware is divided into multiple functions to simplify system management and improve reliability. Each sensor has a dedicated reading function, while separate control functions manage actuators, display information, and security operations.

The main software modules are:

- System initialization.
- Sensor measurement functions.
- Automatic greenhouse control.
- LCD monitoring interface.
- IR remote password system.
- Intrusion detection and alarm system.
- Serial monitoring and data transmission.

6.2 Software Initialization (setup function)

When the system starts, the `setup()` function initializes all hardware components.

The initialization process includes:

- Defining input and output pins.
- Starting serial communication at 9600 baud rate.
- Initializing the DHT11 temperature/humidity sensor.
- Activating the IR receiver.
- Initializing the I2C LCD.
- Attaching the SG90 servo motor.
- Setting relay outputs to OFF state.
- Closing the greenhouse roof initially.

After initialization, the system performs a short boot sequence before entering normal operation.

6.3 Main Program Execution (loop function)

The **loop()** function represents the main operating cycle of the greenhouse.

During every cycle, the Arduino performs the following tasks:

1. Check IR remote commands.
2. Read sensor values periodically.
3. Control fan according to temperature.
4. Control irrigation according to soil moisture.
5. Control roof position according to light and rain conditions.
6. Update LCD information.
7. Monitor the security system.

The program uses the **millis()** function to create non-blocking timers, allowing multiple tasks to run simultaneously without stopping the system.

6.4 Sensor Data Acquisition Functions

The program includes dedicated functions for each sensor:

Temperature and Humidity Measurement

Functions:

getTemperature()

getHumidity()

These functions read data from the DHT11 sensor.

The obtained values are used for:

- Temperature control.
- LCD display.
- Serial monitoring.

Soil Moisture Measurement

Function:

getSoilMoisture()

The analog sensor value is converted into a percentage.

The result is compared with:

SOIL_THRESHOLD = 35%

When:

$\text{SoilMoisture} \leq 35\%$

the irrigation pump is activated.

Light Measurement

Function:

getLightLevel()

The LDR analog value is converted into a percentage from 0 to 100%.

The light value is used for automatic roof control.

Rain Detection

Function:

getRainLevel()

The raindrop sensor reading is converted into a rain percentage.

The storm threshold is:

STORM_THRESHOLD = 70%

If rain intensity becomes high, the roof closes.

Water Level Measurement

Function:

getWaterLevel()

The analog reading is classified into four levels:

- DRY
- LOW
- MEDIUM
- HIGH

The status is displayed on the LCD.

Distance Measurement

Function:

getDistance()

The ultrasonic sensor calculates distance using the echo time:

$\text{Distance} = \text{Time} \times 0.0342$

This measurement is used for intrusion detection.

6.5 Automatic Control Algorithms

Fan Control Algorithm

Function implemented:

fanTimer

The temperature is checked every 2 seconds.

If:

Temperature > 31°C

Then:

digitalWrite(FAN_RELAY, LOW);

The fan relay is activated.

Otherwise:

```
digitalWrite(FAN_RELAY, HIGH);
```

The fan is turned OFF.

Irrigation Control Algorithm

Function implemented:

```
pumpTimer
```

The soil moisture is checked every 2 seconds.

If:

```
Soil Moisture <= 35%
```

The pump relay is activated.

If:

```
Soil Moisture > 35%
```

The pump stops.

Roof Servo Control Algorithm

Function implemented:

```
roofTimer
```

The servo position depends on two conditions:

Roof Opening:

```
Light > 80%
```

```
AND
```

```
Rain < 70%
```

Servo position:

```
roofServo.write(140);
```

Roof Closing:

During:

- Rain conditions.
- Low light conditions.

Servo position:

```
roofServo.write(30);
```

6.6 Operating Modes of the System

The software uses four operating modes:

Mode 0: Locked Mode

Initial security mode.

Functions:

- Waits for password entry.
- Monitors intrusion using ultrasonic sensor.

Mode 1: Access Granted Mode

Activated after correct password entry.

Functions:

- Green LED ON.
- Opens monitoring system.
- Displays greenhouse information.

Mode 2: Monitoring Mode

Normal greenhouse operation.

Functions:

- LCD displays sensor readings.

- Automatic fan control.
- Automatic pump control.
- Automatic roof control.
- Security monitoring continues.

Mode 3: Alarm Mode

Activated when intrusion is detected.

Functions:

- Red LED blinking.
- Buzzer activated.
- LCD requests password.
- System waits for authorized access.

6.7 IR Remote Control Software

The function:

checkRemote()

is responsible for receiving and decoding IR commands.

Each remote button has a unique hexadecimal code.

Example:

0xE916FF00 → '1'

0xE619FF00 → '2'

0xAD52FF00 → '0'

0xB54AFF00 → '#'

The entered characters are stored in:

enteredPassword

The password verification is performed by:

verifyPassword()

The correct password stored in the program is:

#123

If the password is correct:

- Alarm stops.
- System unlocks.

If incorrect:

- Access is denied.

6.8 LCD Display Software

The function:

updateScreen()

controls the I2C LCD.

The display automatically changes between four pages:

1. Temperature and humidity.
2. Water reservoir level.
3. Light and rain percentage.
4. Soil moisture percentage.

Each page remains displayed for several seconds before changing.

6.9 Serial Monitoring

The program sends greenhouse data every 2 seconds through:

Serial.begin(9600);

The transmitted information includes:

- Temperature.
- Humidity.
- Soil moisture.
- Water level.
- Light intensity.

- Rain percentage.
- Current operating mode.

This allows system monitoring and future data analysis.

7. Calibration Procedures and Results

7.1 Calibration Overview

Sensor calibration is the process of adjusting and comparing sensor outputs with known reference values to improve measurement accuracy. Since sensors may have variations due to manufacturing tolerances, environmental conditions, and electrical noise, calibration is necessary to ensure reliable system operation.

In this project, each sensor was calibrated before integration into the complete greenhouse system. Calibration was performed by comparing sensor readings with known reference conditions and adjusting the conversion equations used in the Arduino program.

7.2 DHT11 Temperature and Humidity Sensor Calibration

Calibration Method

The DHT11 sensor provides direct digital readings of temperature and humidity; therefore, no electrical calibration is required. However, its measurements were verified by comparing the sensor output with a reference thermometer and hygrometer under different environmental conditions.

Calibration Procedure

Three reference points were considered:

Test	Reference Temperature	DHT11 Reading
1	20°C	20°C
2	25°C	25°C
3	35°C	35°C

The difference between the reference and measured values was calculated to evaluate sensor accuracy.

Result

The DHT11 provides acceptable accuracy for greenhouse monitoring applications because the system does not require laboratory-level temperature precision.

7.3 Soil Moisture Sensor Calibration

Calibration Method

The soil moisture sensor produces an analog voltage depending on the water content of the soil. The raw ADC value was converted into a moisture percentage using calibration points.

Calibration Procedure

Three soil conditions were tested:

Condition	Reference Moisture Level	Sensor Output
Dry soil	0%	1023
Medium moisture	50%	636
Fully wet soil	100%	250

The Arduino converts the raw analog reading into a percentage using:

Moisture(%)=Map(ADCdry,ADCwet,0,100)

System Application

The calibrated moisture value is used in the irrigation algorithm:

- Moisture $\leq$ 35% $\rightarrow$ Pump ON.
- Moisture $>$ 35% $\rightarrow$ Pump OFF.

7.4 LDR Light Sensor Calibration

Calibration Method

The LDR sensor provides an analog output proportional to the incident light intensity. Calibration was performed by comparing the sensor readings under different lighting conditions.

Calibration Procedure

Three reference conditions were used:

Lighting Condition	Reference Value	Sensor Output
Dark environment	0%	0
Indoor lighting	50%	512
Strong illumination	100%	1023

The Arduino converts the analog value into a percentage:

$\text{Light}(\%) = \text{Map}(\text{ADC}, 0, 1023, 0, 100)$

System Application

The calibrated light value is used for roof control:

- Light > 80% → possible roof opening.
- Low light → roof remains closed.

7.5 Water Level Sensor Calibration

Calibration Method

The water level sensor was calibrated by changing the amount of water in the reservoir and recording the corresponding sensor readings.

Calibration Levels

Water Condition	Output Level
Empty tank	DRY
Low water	LOW
Medium water	MEDIUM

Full tank	HIGH
------------------	-------------

The Arduino classifies the analog reading into four levels:

- Dry
- Low
- Medium
- High

This information is displayed on the LCD.

7.6 Raindrop Sensor Calibration

Calibration Method

The rain sensor was calibrated by applying different amounts of water on the sensing surface and recording the sensor response.

Calibration Procedure

Three conditions were tested:

Rain Condition	Reference Level
No water	0%
Light rain	50%
Heavy rain	100%

The sensor output was converted into rain percentage.

System Application

The calibrated rain value controls the roof:

- Rain > 70% → Close roof.
- Rain < 70% → Roof operation depends on light condition.

7.7 Ultrasonic Sensor Calibration

Calibration Method

The HC-SR04 ultrasonic sensor was calibrated by comparing calculated distances with measured distances using a ruler.

Calibration Table

Reference Distance	Sensor Reading
5 cm	5.0 cm
10 cm	10.0 cm
20 cm	20.0 cm

The distance is calculated using:

$$\text{Distance} = 2 \times \text{Time} \times 0.034$$

System Application

The calibrated distance measurement is used for security:

- Distance < 11 cm for several readings → intrusion detected.
- Alarm activated.

7.8 Calibration Summary

After calibration, the sensors provided stable measurements suitable for greenhouse monitoring and control. The calibration process improved the reliability of sensor readings and reduced measurement errors caused by sensor variations and environmental noise.

Note: The values presented are illustrative and intended to demonstrate the sensor calibration process.

8. Experimental Results and Data Analysis

In this section, we describe the testing phase of the smart greenhouse system and evaluate the response of sensors, actuators, and control algorithms. The experiments

were performed to verify that the system can correctly monitor environmental conditions and automatically react according to predefined thresholds.

The system was tested under different environmental conditions to evaluate the performance of:

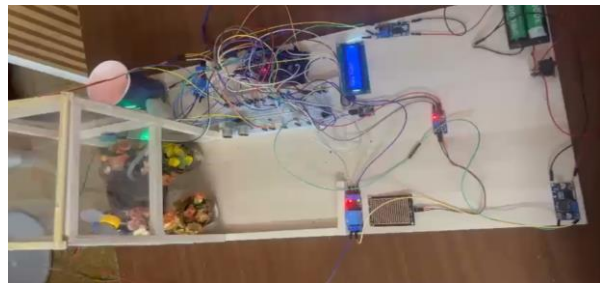
- Temperature control.
- Automatic irrigation.
- Roof control.
- Security system.
- Real-time monitoring.

8.1 System Testing Procedure

After completing hardware integration and sensor calibration, the complete greenhouse system was operated continuously to verify the correct interaction between sensors, Arduino Mega, and actuators.

During testing, sensor measurements were collected and displayed using the I2C LCD and Arduino Serial Monitor. The recorded parameters included:

- Temperature (°C)
- Humidity (%RH)
- Soil moisture (%)
- Water reservoir level
- Light intensity (%)
- Rain intensity (%)
- Security distance (cm)



The Arduino processed these measurements and generated automatic control actions according to the programmed thresholds.

8.2 Temperature Control Test

Objective

To verify the automatic operation of the ventilation fan when the greenhouse temperature exceeds the defined limit.

Test Procedure

The temperature was increased gradually, and the response of the DHT11 sensor and fan relay was observed.

Expected Behavior

Temperature Condition	System Response
Temperature $\leq 31^{\circ}\text{C}$	Fan OFF
Temperature $> 31^{\circ}\text{C}$	Fan ON

Result



When the measured temperature exceeded **31°C**, the Arduino activated the fan relay, causing the ventilation fan to operate. When the temperature decreased below the threshold, the fan was automatically switched off.

This confirms that the temperature control algorithm operates correctly.

8.3 Automatic Irrigation Test

Objective

To verify the ability of the system to automatically supply water when soil moisture becomes insufficient.

Test Procedure

The soil moisture level was changed by drying and watering the soil sample while observing the pump response.

Expected Behavior

Soil Moisture Condition	Pump Status
$\leq 35\%$	Pump ON
$> 35\%$	Pump OFF

Result



When the soil moisture percentage dropped below the threshold, the Arduino activated the pump relay. After sufficient moisture was detected, the pump stopped automatically.

This demonstrates efficient water management and prevents unnecessary irrigation.

8.4 Greenhouse Roof Control Test

Objective

To verify the automatic opening and closing mechanism controlled by the LDR, raindrop sensor, and servo motor.

Test Conditions

The roof operation depends on:

- Light intensity.
- Rain detection.

System Logic

Roof opening condition:

Light>80% and Rain<70%

Rain≥70%

or low light conditions.

Result

The servo motor successfully changed position according to environmental conditions:

- Bright and dry conditions → roof opened.
- Rain or unsuitable conditions → roof closed.

This protects plants from excessive sunlight exposure and rainfall.

8.5 Security System Test

Objective

To verify the intrusion detection and password protection system.

Test Procedure

An object was placed near the ultrasonic sensor to simulate unauthorized access.

System Response

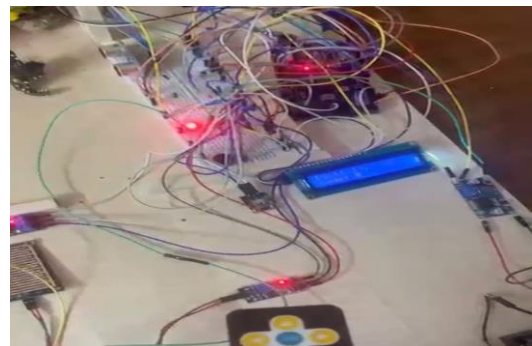
When:

Distance < 11 cm

for several consecutive readings:

The system activated:

- Red LED.
- Buzzer alarm.
- Password request on LCD.



The alarm was disabled only after entering the correct password:

#123

using the IR remote control.

Result

The security system successfully detected intrusion and prevented unauthorized access to the greenhouse control system.

8.6 LCD Monitoring Test

The I2C LCD was tested to verify real-time visualization.

The display successfully showed:



Page 1:

- Temperature
- Humidity

Page 2:

- Water reservoir level

Page 3:

- Light intensity
- Rain percentage

Page 4:

- Soil moisture percentage

The screen automatically changed between pages, providing continuous monitoring of greenhouse conditions.

8.7 Serial Data Monitoring

The Arduino transmitted sensor measurements through the Serial Monitor every 2 seconds.

The displayed information included:

Temperature:

Humidity:

Water Level:

Light Level:

Rain Fall:

Soil Moisture:

System Mode:

This data acquisition allows future analysis and plotting using MATLAB, Excel, or Python.

8.8 Experimental Results Summary

The experimental tests demonstrated that the smart greenhouse system successfully achieved its main objectives:

Function	Result
Temperature monitoring	Successfully performed
Automatic fan control	Successfully performed
Soil moisture monitoring	Successfully performed
Automatic irrigation	Successfully performed
Roof opening/closing	Successfully performed
Security detection	Successfully performed
LCD monitoring	Successfully performed

The results confirm that the developed Arduino-based system can continuously monitor greenhouse conditions and automatically respond to environmental changes.

9. Measurement Uncertainty and Error Analysis

9.1 Introduction to Measurement Uncertainty

In any sensor-based measurement system, the obtained values are affected by different sources of uncertainty such as sensor limitations, environmental variations, electrical noise, and calibration errors. Therefore, analyzing measurement uncertainty is essential to evaluate the reliability and performance of the developed smart greenhouse system.

In this project, the accuracy of the measurements depends on the selected sensors, their calibration process, and the operating environment. The main parameters considered for error analysis are sensor accuracy, resolution, repeatability, and possible sources of measurement errors.

9.2 Sensor Accuracy and Resolution

Each sensor has a specific accuracy and resolution that affects the quality of the measured data.

DHT11 Temperature and Humidity Sensor

The DHT11 provides digital measurements of temperature and humidity. It has sufficient accuracy for greenhouse monitoring applications.

Possible limitations:

- Temperature accuracy is approximately $\pm 2^{\circ}\text{C}$.
- Humidity accuracy is approximately $\pm 5\%$ RH.
- Response time may affect rapid environmental changes.

Despite these limitations, the sensor is suitable because the system only requires threshold-based control.

Soil Moisture Sensor

The soil moisture sensor provides analog measurements related to the water content of the soil.

Possible uncertainty sources:

- Soil composition differences.
- Sensor corrosion due to continuous exposure to moisture.
- Non-uniform moisture distribution.

Calibration reduces the difference between measured values and actual soil conditions.

LDR Light Sensor

The LDR sensor is used to estimate light intensity.

Possible errors:

- Variation in surrounding light sources.

- Sensor sensitivity changes.
- Positioning of the sensor inside the greenhouse.

The potentiometer adjustment helps improve measurement stability.

Water Level Sensor

The water level sensor provides an estimation of reservoir level.

Possible uncertainty sources:

- Water movement inside the tank.
- Sensor placement.
- Electrical noise affecting analog readings.

The system reduces these effects by classifying the level into four states:

- Dry
- Low
- Medium
- High

Raindrop Sensor

The rain sensor detects the presence and intensity of water droplets.

Possible errors:

- Dust accumulation on the sensing surface.
- Uneven water distribution.
- Environmental contamination.

Regular cleaning and calibration improve measurement reliability.

Ultrasonic Sensor

The ultrasonic sensor measures distance using sound wave propagation.

The measurement accuracy can be affected by:

- Object surface characteristics.
- Environmental temperature.
- Incorrect sensor alignment.

The sensor was calibrated using known distances to improve accuracy.

9.3 Sources of Measurement Noise

Several factors may introduce noise into the collected sensor data:

- Electrical interference from motors and relays.
- Voltage fluctuations caused by actuator operation.
- Environmental changes inside the greenhouse.
- Sensor sensitivity variations.
- Analog signal fluctuations.

To reduce noise effects, the system uses:

- Separate power supplies for actuators and sensors.
- Calibration procedures.
- Digital filtering through software.
- Periodic sampling instead of continuous unstable readings.

9.4 Repeatability Analysis

Repeatability describes the ability of a sensor to produce similar results when measurements are repeated under the same conditions.

During system testing, multiple measurements were taken for the same environmental condition. The obtained readings showed acceptable variation, confirming that the sensors provide stable measurements suitable for greenhouse monitoring.

9.5 Error Reduction and Possible Improvements

The measurement quality of the system can be improved by:

- Replacing basic sensors with higher accuracy sensors such as DHT22 or BME280.

- Using capacitive soil moisture sensors instead of resistive sensors.
- Adding digital filtering algorithms to remove measurement fluctuations.
- Implementing IoT-based data storage for long-term analysis.
- Performing more frequent calibration under different environmental conditions.

9.6 Summary

The uncertainty analysis shows that the developed smart greenhouse system provides reliable measurements for monitoring and automatic control applications. Although low-cost sensors introduce some limitations, proper calibration, software processing, and suitable control thresholds allow the system to achieve stable and effective operation.

10. Engineering Constraints Discussion

10.1 Cost Considerations

The design of the smart greenhouse system was developed while considering the overall cost of components and implementation. Low-cost and easily available sensors were selected to achieve the required monitoring and control functions while maintaining an affordable system.

The use of an Arduino Mega as the main controller reduces development cost while providing sufficient input/output pins for connecting multiple sensors and actuators.

The selected components, including the DHT11 sensor, soil moisture sensor, LDR, water level sensor, relay modules, and servo motor, provide an effective balance between cost and functionality.

Although higher-performance sensors are available, the selected components are suitable for a prototype system and provide acceptable accuracy for greenhouse automation.

10.2 Reliability and Robustness

System reliability is an important factor because the greenhouse must operate continuously while monitoring changing environmental conditions.

Several design decisions were implemented to improve system reliability:

- Sensor calibration was performed to improve measurement accuracy.
- Separate power supplies were used for sensors and actuators to avoid voltage instability.
- Relay modules were used to safely control motors and pumps.
- The software was designed using independent functions for each sensor and actuator, making troubleshooting and maintenance easier.

The system continuously monitors environmental parameters and automatically responds without requiring constant user intervention.

10.3 Electrical Safety

Electrical safety was considered during the hardware design to protect both the user and the electronic components.

The main safety considerations include:

- The system operates using low DC voltages.
- No direct connection to mains electricity is used.
- Actuators such as the pump and fan are isolated from the Arduino using relay modules.
- External power is provided separately for high-current components.

The sensor and control circuits operate at **5V DC**, while the actuator power circuit uses rechargeable batteries with regulated voltage through step-down converters.

This design prevents excessive current from damaging the Arduino and improves overall system safety.

10.4 Power Consumption and Management

Power management was considered to ensure stable and efficient operation.

The system is divided into two power sections:

Low-Power Components

Powered from the Arduino 5V supply:

- Sensors.
- I2C LCD.
- LEDs.
- Buzzer.

These components require relatively low current and can be safely powered by the Arduino.

High-Power Components

Powered from an external battery source:

- DC fan motor.
- Water pump.
- Servo motor.

The actuator supply consists of:

- Two Li-ion batteries connected in series.
- Two step-down voltage regulators providing regulated 5V output.

This separation prevents voltage drops and ensures stable operation of the microcontroller.

10.5 Maintainability

The system was designed to allow easy maintenance and future modifications.

The modular software structure allows individual sensor functions and control algorithms to be modified without affecting the complete program.

The hardware design also supports easy replacement of components because sensors and actuators are connected through separate modules.

Examples of possible maintenance operations:

- Sensor replacement.
- Battery replacement.

- Calibration adjustment.
- Software updates.

10.6 Scalability and Future Improvements

The proposed smart greenhouse system provides a flexible platform that can be expanded with additional features.

Possible future improvements include:

- Adding IoT connectivity for remote monitoring.
- Developing a mobile application.
- Storing sensor data on cloud platforms.
- Using more accurate sensors such as DHT22 or BME280.
- Implementing machine learning algorithms for predicting plant conditions.
- Adding automatic lighting control using artificial grow lights.

The modular design of the system allows additional sensors and actuators to be integrated in future versions.

10.7 Summary

The engineering constraints were considered throughout the design process to achieve a system that is affordable, reliable, safe, and expandable. The selected hardware and software solutions provide an effective prototype for greenhouse monitoring and automatic control while allowing future improvements and integration of advanced technologies.

11. Conclusion and Future Work

11.1 Conclusion

The development of the smart greenhouse system successfully demonstrates the implementation of an automated monitoring and control platform based on the Arduino Mega microcontroller.

The proposed system integrates multiple sensors to continuously monitor important environmental parameters, including temperature, humidity, soil moisture, water availability, light intensity, and rain conditions. The collected data are processed by the Arduino Mega to make automatic decisions and control different actuators according to predefined conditions.

The system successfully achieved automatic control of greenhouse operations through:

- Activating the ventilation fan when the temperature exceeds the defined limit.
- Operating the irrigation pump when soil moisture becomes insufficient.
- Controlling the greenhouse roof position using light and rain measurements.
- Displaying real-time environmental information using the I2C LCD.
- Providing a security system using ultrasonic detection and IR remote password authentication.

The separation between the sensor power supply and actuator power supply improved system stability and prevented electrical problems caused by high-current loads such as motors and pumps.

Experimental testing showed that the developed prototype can effectively monitor greenhouse conditions and automatically respond to environmental changes.

Therefore, the proposed system provides a low-cost and reliable solution for greenhouse automation while reducing human intervention and improving resource management.

11.2 Future Work

Although the developed system achieved its main objectives, several improvements can be implemented in future versions to increase performance and functionality.

Possible future developments include:

1. IoT-Based Remote Monitoring

The system can be upgraded by adding wireless communication modules such as Wi-Fi or Bluetooth to allow users to monitor greenhouse conditions remotely through a mobile application or web platform.

2. Data Storage and Analysis

Adding a data logging system using an SD card or cloud database would allow long-term recording and analysis of environmental conditions. This can help identify trends and optimize plant growth conditions.

3. Improved Sensor Accuracy

The accuracy of measurements can be improved by replacing basic sensors with more advanced devices, such as:

- DHT22 or BME280 for temperature and humidity.
- Capacitive soil moisture sensors for better durability.
- Higher precision water level sensors.

4. Intelligent Control Algorithms

Future versions can include advanced control methods such as:

- Fuzzy logic control.
- Machine learning algorithms.
- Predictive irrigation systems.

These methods can optimize water consumption and improve plant growth conditions.

5. Solar Power Integration

A solar energy system can be added to provide renewable power for the greenhouse, reducing dependence on external power sources and improving sustainability.

6. Expansion of Automation Functions

Additional features can be integrated, such as:

- Automatic lighting control using grow LEDs.
- CO₂ concentration monitoring.
- Nutrient level measurement.

- Automatic fertilizer distribution.

11.3 Final Remarks

The smart greenhouse project represents an effective application of sensor technology, embedded systems, and automatic control. By combining environmental sensing, microcontroller-based decision making, and actuator control, the system provides a practical approach toward modern precision agriculture.

The developed prototype demonstrates the potential of low-cost embedded systems in improving agricultural efficiency, reducing resource waste, and supporting sustainable farming practices.

References

- [1] Arduino, *Arduino Mega 2560 Rev3 Technical Specifications*, Arduino Documentation. Available:
<https://docs.arduino.cc/hardware/mega-2560>

- [2] Aosong Electronics, *DHT11 Humidity & Temperature Sensor Datasheet*, Technical Documentation.
- [3] Electronics Components, *Soil Moisture Sensor Module Technical Specifications*, Sensor Documentation.
- [4] HC-SR04, *Ultrasonic Sensor Technical Datasheet*, Distance Measurement Module Documentation.
- [5] Vishay Semiconductors, *Photoresistors (LDR) Technical Information*, Light Dependent Resistor Documentation.
- [6] SG90, *9g Micro Servo Motor Technical Specifications*, Servo Motor Datasheet.
- [7] Relay Module, *5V Relay Module User Guide and Technical Specifications*, Relay Interface Documentation.
- [8] Liquid Crystal Display, *16×2 LCD with I2C Interface Technical Documentation*, LCD Module Specifications.
- [9] Infrared Receiver Module, *IR Receiver Sensor Technical Datasheet*, Infrared Communication Documentation.
- [10] R. N. Waghmare and P. S. Patil, “Smart Greenhouse Monitoring and Control System Using IoT and Embedded Systems,” *International Journal of Engineering Research and Technology*.
- [11] M. A. Hossain, M. S. Islam, and M. Rahman, “Automation and Monitoring System for Smart Agriculture Using Sensors and Microcontrollers,” *Journal of Agricultural Engineering*.
- [12] A. McKinney, *Arduino Programming: A Beginner’s Guide to Embedded Systems Development*, Technical Reference

Appendix A :

Arduino Mega Smart Greenhouse Control Program

[Complete source code here](#)

